Poulson Prints ERP — System Spec

Nov 11, 2026 · @Ryan

Company overview

Poulson Prints is a fictional Utah 3D-printing business that sells mostly online, with a campus pickup counter at BYU. It runs 4 Bambu Lab P2S printers (each with an AMS multi-color unit) and follows a pull model: popular items are kept in stock, everything else is printed to order.

Location	Role
Stansbury Park Print Farm	Printers, filament, raw materials, work in progress
Stansbury Park Fulfillment Center	Finished goods, packaging, packing station, shipping
BYU Campus Pickup Counter	Walk-up POS sales, display stock, pickup orders
QA Hold (quarantine area inside the Print Farm)	Returned quality failures awaiting inspection, kept apart from sellable stock on its own 8 cu ft shelf

- **Staff hours:** 8am to 8pm. Prints can run past 8pm, but completion waits until 8am the next morning.
- **Labor rate:** $20/hr. Setup 1 min and cleanup 1 min per print (per build plate, not per unit).
- **Delivery runs:** Tuesday and Friday, Stansbury Park to BYU, about 65 miles each way.
- **Brand:** navy with teal accent; logo is a "P" built from stacked print layers with a nozzle printing the top layer.
- **Goal:** meet all 12 required ERP capabilities from the GSCM 401 rubric, add wow factors that would genuinely help the business, and make every number traceable to stored data.

Products and prices

Nine products become 30 sellable items: single-color products come in red, white, blue or black, and each color is its own item with its own barcode. Prices are final (the lower of our price and the market recommendation); margins recalculate live as costs change.

Product	Variants	Stocking	Unit cost	Price	Margin	Size (in)	Box	Packing class
Dragon	4 colors	In stock	$1.07	$8.99	88%	4 × 2 × 1.5	Small	Standard (1 min)
Keychain (blue/white, printed in lots of 10)	1	In stock	$0.93	$4.99	81%	2 × 1 × 0.25	Small	Small (0.5 min)
Fidget Spinner	4 colors	In stock	$2.85	$7.99	64%	3 × 3 × 0.5	Small	Standard (1 min)
Christmas Ornament (seasonal Nov 1 – Dec 31)	4 colors	In stock in season	$1.44	$7.99	82%	3 × 3 × 0.5	Small	Standard (1 min)
Temple (LED tealight)	4 colors	Made to order	$6.65	$27.99	76%	4 × 4 × 5	Medium	Fragile (2 min)
Army Soldiers (lot of 10, sold only as a lot)	4 colors	Made to order	$9.98	$19.99	50%	5 × 3 × 2	Small	Multi-piece (1.5 min)
Airplane (black/blue/white/red)	1	Made to order	$10.37	$39.99	74%	10 × 9 × 3	Large	Fragile (2 min)
BYU Logo	4 colors	Made to order	$2.31	$14.99	85%	7 × 5 × 0.75	Medium	Standard (1 min)
Magnet Alphabet Set (26 letters)	4 colors	Made to order	$10.69	$22.99	54%	5 × 4 × 1.5	Small	Multi-piece (1.5 min)

- **Unit cost** = product cost only (filament + machine time + labor + components). Packing labor, packaging and postage are fulfillment cost, tracked per order.
- **Units per plate:** default 1 for every product, editable per product. Hints shown in the app: Dragon 6, Spinner 4, Ornament 8.
- **Multi-color purge waste:** a setting (grams per color change) adds AMS purge filament to the Keychain and Airplane cost.
- **Seasonal ornament:** shown in the customer portal only inside its window; forecast as "sells like the Dragon × 1.5" until it has its own history.
- **New designs:** added through the New Design wizard (see Wow factors).

Bills of materials and cost roll-up

Every product has a single-level bill of materials (9 parents, above the rubric's 5). Issuing a production order consumes these components; completing it receives the finished item at the rolled-up cost of what was actually used.

Product (per plate)	Print time	Filament	Components	Assembly	Filament $	Machine $	Labor $	Parts $	Total cost
Temple (1)	5 h 5 min	58.36 g, chosen color	1 g glue, 1 LED tealight	10 min	$0.89	$1.07	$4.00	$0.70	$6.65
Dragon (1)	1 h 9 min	10.43 g, chosen color	none	0 min	$0.16	$0.24	$0.67	$0.00	$1.07
Keychain (lot of 10)	3 h 8 min	13.96 g blue + 2.47 g white	10 split rings, 10 chains	2 min each (20 min)	$0.25	$0.66	$7.33	$1.10	$9.34 lot / $0.93 each
Army Soldiers (lot of 10)	3 h 0 min	27.7 g, chosen color	2 g glue, 10 figure bases	20 min per lot	$0.42	$0.63	$7.33	$1.60	$9.98
Airplane (1)	7 h 49 min	14.77 g black, 55.59 g blue, 84.99 g white, 16.29 g red	1.5 g glue	15 min	$2.61	$1.64	$5.67	$0.45	$10.37
BYU Logo (1)	1 h 44 min	84.44 g, chosen color	none	0 min	$1.28	$0.36	$0.67	$0.00	$2.31
Magnet Alphabet Set (1)	1 h 59 min	41.52 g, chosen color	26 magnets, 2.5 g glue	20 min	$0.63	$0.42	$7.33	$2.31	$10.69
Fidget Spinner (1)	45 min	18 g, chosen color	1 608ZZ bearing, 3 hex nuts	3 min	$0.27	$0.16	$1.67	$0.75	$2.85
Christmas Ornament (1)	50 min	12 g, chosen color	1 ribbon hanger	1 min	$0.18	$0.18	$1.00	$0.08	$1.44

Labor includes 1 min setup + 1 min cleanup per plate plus assembly. Costs in the app are calculated live from current settings and weighted-average component costs, so these figures are the starting point, not stored values.

$$\text{Unit cost} = \frac{g \cdot c_{g} + h \cdot r_{m} + \frac{(2 + a)}{60} \cdot r_{l} + \sum \text{component costs}}{\text{units per plate}}$$

where g = filament grams, c_g = $0.0152/g, h = print hours, r_m = $0.21/hr machine rate, a = assembly minutes, r_l = $20/hr labor.

Purchased items and suppliers

Seven real suppliers provide 17 purchased items. Lead times, terms and prices are realistic assumptions, contact names are fictional, and scorecard performance comes from simulated purchase history (the scorecard screen says so).

Supplier	Items	Unit cost	Quoted lead time	Terms	Website
Hobby Lobby	PLA filament red, white, blue, black (985 g spool, part 2589638); ribbon hanger	$14.99/spool; $0.08	2 days	Paid at purchase	hobbylobby.com
Bambu Lab	PLA filament, backup source (1 kg)	$19.99/spool	7 days	Paid at purchase	bambulab.com
Uline	Boxes Small 6×4×4, Medium 9×7×5, Large 12×10×6; tissue paper	$0.45, $0.75, $1.25; $0.05/sheet	2 days	Net 30	uline.com
Amazon Business	Super glue (per g), LED tealight, split ring, keychain chain, figure base 25 mm, hex nut	$0.30, $0.40, $0.06, $0.05, $0.10, $0.05	3 days	Net 30	business.amazon.com
K&J Magnetics	Neodymium magnet 6×2 mm	$0.06	6 days	Net 15	kjmagnetics.com
VXB Bearings	608ZZ bearing	$0.60	8 days	Net 30	vxb.com
Vistaprint	Thank-you card	$0.15	10 days	Paid at purchase	vistaprint.com

- **Item count:** 30 finished items + 17 purchased items = 47 (rubric minimum 25).
- **Each spool is its own barcode** and is tracked down to the gram (starts at 985 g).
- **Purchase prices vary** in the history (Hobby Lobby sales), so weighted-average cost moves realistically.
- **Two filament sources** let the scorecard compare price, lead time and reliability side by side.
- **Packaging rule:** box and tissue per box shipped, 1 thank-you card per order.

Inventory, capacity and holding costs

Inventory is never typed in: on-hand quantities, spool weights and values are calculated from the transaction history (receipts, production, sales, returns, transfers, adjustments).

Capacity by location

Location	Capacity	Warning levels
Print Farm	48 cu ft shelving (40 cu ft general + 8 cu ft QA Hold quarantine) + filament rack of 60 spools	Yellow above 85%, red above 100%
Fulfillment Center	120 cu ft	Yellow above 85%, red above 100%
Campus Pickup Counter	12 cu ft	Yellow above 85%, red above 100%

QA Hold has its own 8 cu ft capacity, tracked separately from the Print Farm's general shelving so quarantined items never take space counted as sellable stock. When QA Hold passes 85%, the dashboard flags an inspection backlog: items need to be restocked or scrapped.

- Utilization = item dimensions × quantity on hand (boxes counted folded flat).
- Each item can also have a max stock level per location.
- **Over capacity = warn, never block.** An ERP must record what physically happened. Over 100% needs a manager override with a reason code, logged with who, when and why.
- When an override is used, the system suggests a fix: transfer to a location with room, split the delivery, or pull production forward.

- Returns, completed production for existing orders and goods already received are always allowed, but flagged.
- EOQ, reorder and MRP suggestions are capped by available space ("EOQ says 18 spools, rack has room for 11").

Holding cost rates (all editable)

Component	Rate	Bucket
Cost of capital	10% per year	Opportunity
Insurance	1% per year	Cash
Expected shrinkage	2% per year	Opportunity (estimate)
Expected obsolescence	5% per year; 20% for seasonal items	Opportunity (estimate)
Storage space	$1.50 per cu ft per month	Cash

Total carrying rate: 18% per year for standard items (filament included), 33% for seasonal items, plus space cost.

Holding cost report: three buckets

1. **Cash costs** — storage rent and insurance. On the P&L as operating expense.
2. **Realized losses** — actual shrinkage, damage and obsolescence write-offs, calculated from inventory adjustments with reason codes. On the P&L.
3. **Opportunity and risk costs** — cost of capital and expected shrinkage/obsolescence. Management report only; feeds EOQ.

The report compares expected shrinkage (2%) to actual adjustments, flags slow-moving stock (no sales in 60 days), and shows leftover seasonal stock after its window closes.

The 12 required capabilities

Each rubric capability maps to screens in the app, and every one posts to the same shared database so the modules stay connected.

#	Capability	What the system does	Connects to
1	Point of Sale	Multi-line sales at the campus counter; scan to add; Utah sales tax; discounts (BYU student 10%, business 15% on 25+ units, manual with manager reason; one discount per order, best applies); printed/emailed receipt with tracking QR code	Inventory (reduces stock), ledger (revenue, discounts, tax, COGS), customer tracking email
2	Inventory Management	On-hand by item and location (3 + QA Hold); transfers; adjustments with reason codes (cycle count, damaged, lost, obsolete); backorders with estimated available date; capacity and holding costs	Every module; adjustments feed realized losses on the P&L
3	Barcode Support	Code 128 label on every item and every filament spool; printable label sheet; phone-camera scanning (typed entry as backup) for scan-to-add in POS and scan-to-receive/adjust/transfer/count in inventory	POS, receiving, cycle counts, spool loading
4	Sales Orders	Customer, location, items, quantities; status Order Placed → Fulfilled → Closed/Cancelled, with detailed tracking stages underneath (Queued, Printing, Assembly, Packing, Shipped/Ready for Pickup)	MRP demand, printer queue, box selection, shipping, tracking email
5	Returns and Exchanges	30-day policy; exchange for different color/product; quality failures get a reason code, full refund, and go to QA Hold; "changed my mind" on made-to-order items has a 10% restocking fee; revenue and COGS reversed or adjusted	Inventory, ledger, failure analytics

#	Capability	What the system does	Connects to
6	Production Orders	Item, quantity, location; issue consumes BOM components (filament from specific spools); complete receives finished goods at rolled-up cost; records qty started, good, scrapped/reworked with reason code, start and finish times; Report Failure button; metrics: first-pass yield, scrap rate, average cycle time, on-time completion, printer utilization	Inventory, spool tracking, failure analytics, order tracking; production report
7	Purchase Orders and Suppliers	Supplier records with contact, quoted lead time, payment terms; POs with status Order Sent → Received → Closed; expected receipt = sent date + lead time; partial receipts into a chosen location; reorder points, periodic review screen, EOQ from sales history; supplier scorecard with overall rating	Inventory, AP, MRP, capacity
8	MRP	Confirmed sales orders × BOM = component requirements, minus on-hand and open POs = suggested order quantity, with order-by date = need date − lead time; 4-week weekly view; one click to create the PO	Sales orders, BOMs, inventory, POs
9	Financial Reporting	P&L (gross sales, discounts, net sales, shipping revenue, COGS, fulfillment cost, cost of poor quality, operating expenses); inventory valuation at weighted average cost; inventory turnover; all live from the ledger	Every transaction

#	Capability	What the system does	Connects to
10	Reporting	Sales by day/week, top sellers, low stock with suggested order qty and order-by date, inventory by location, supplier performance, production report, holding cost report, failure analytics	All modules
11	Customer-Facing ERP	Customer login; available inventory by location; place pickup or shipped orders; out-of-stock items still orderable with a delivered-by date; quote calculator; order tracking	Sales orders, inventory, printer queue
12	Multi-user logins	3 identical staff logins + 1 demo customer login; Supabase Realtime pushes changes to every open screen instantly; live activity feed	Everything

Key formulas (shown live in "Show the math" panels)

$$\text{Reorder point} = \bar{d} \times L + z \sigma_d \sqrt{L}$$

$$EOQ = \sqrt{\frac{2DS}{H}}$$

$$\text{Inventory turnover} = \frac{\text{COGS (period)}}{\text{average inventory value}}$$

- $\bar{d}$ = average daily usage from sales history, σd = its standard deviation, L = lead time in days, z = service-level factor (setting, default 1.65 for 95%).
- D = annual demand from sales history, S = ordering cost per order (setting, default $15), H = annual holding cost per unit (from the holding cost rates).
- For finished goods made in-house, the "lead time" is production lead time and EOQ becomes the economic production batch size; S is the setup cost.
- Supplier scorecard shows on-time delivery rate, average actual vs. quoted lead time, fill rate, and payment tracking (amounts owed, due dates from terms), all from PO history. Overall rating = weighted score of on-time rate (40%), fill rate (30%), lead time accuracy (20%) and price stability (10%) → Keep (80+), Watch (60–79), Replace (below 60).

Wow factors and extra features

The five team wow factors all run on data the ERP already tracks, so each one also demonstrates how the modules connect.

1. Live order tracking with email link

- After any order (online, quote, POS), the system asks for an email and sends a "Track your order" link via EmailJS from a team Gmail account.
- The link is private and unguessable; no login needed.
- Tracker shows the stage, % printed, and a countdown to ready/delivered, calculated from print time, staff hours and the printer queue.
- **Demo safety net:** the confirmation screen also shows the link and a QR code; every email is logged as sent or failed.

2. Custom quote calculator

- Customer enters design name/description, estimated print hours, grams and color, and chooses shipping or BYU pickup. An optional file attachment is stored but not analyzed.
- Price = filament + machine time + labor + packaging + failure allowance (from actual failure history) × markup, minimum $10, plus shipping and tax.
- Ready date = next open printer slot + print time + assembly + packing + transit or next delivery run.
- One click turns an accepted quote into a sales order.

3. Automatic box selection

- Packs items largest first into the fewest, smallest boxes (80% fill limit, longest-side check).
- Packing slip explains why each box was chosen; box choice drives packing time, packaging cost and postage.

4. Live filament spool tracking

- Every spool has its own barcode, starting at 985 g; scan to load it onto a printer's AMS slot.
- Grams are deducted as prints run and when failures waste filament; spools near empty are flagged; filament runout is a failure reason.
- A printer board shows each P2S, its loaded spools, current job and time remaining.

5. Print failure analytics

- Reasons: warping, bed adhesion, spaghetti/detached, layer shift, nozzle clog, filament runout, stringing/quality reject, power/network loss.
- Failure rate by printer, product, color and reason; wasted grams and dollars; trends over time.
- Seeded patterns: one printer due for maintenance, the tall Airplane warps more, runouts near spool ends.
- **Report Failure** button on a running print writes off wasted filament, records scrap with reason, auto-queues a reprint and pushes back the customer's tracking date. An optional auto-simulate toggle (off by default) creates random failures at historical rates.

Also included

- **New Design wizard** — add seasonal or new products: print data, components, size, colors, stocking policy; generates items, barcodes, BOM, cost and suggested price. Lifecycle Draft → Active → Seasonal → Retired; "like item" forecasting for designs with no history.
- **Demand simulator** — "Simulate next day/week" generates new orders from the demand model, with adjustable events (e.g. BYU game this Saturday).
- **Shipping estimator** — USPS Ground Advantage–style rate table by zone (from Stansbury Park) and weight; shipping revenue vs. postage expense on the P&L; free BYU pickup with Tuesday/Friday delivery runs at about $0.70/mile.
- **Show the math** panel on every calculated number.
- **QA Hold** location for quality returns.
- **Cost of poor quality** line on the P&L.
- **Accounts receivable and payable** with payment recording.
- **Live activity feed** proving multi-user updates.
- **Requirements map** page linking every rubric line to its screen, plus the required documentation of AI tools and database.
- **Reset demo data** button.

Settings

Every assumption below is an editable setting stored in the database; changing one updates costs, quotes, EOQ, MRP and reports everywhere.

Area	Setting	Default
Labor	Labor rate	$20/hr
Labor	Setup / cleanup per plate	1 min / 1 min
Labor	Staff hours	8am–8pm daily
Machine	Printer purchase cost / life	~$700 / 5,000 print hours
Machine	Electricity	0.15 kW × ~$0.11/kWh
Machine	Maintenance	$0.05/hr
Machine	Machine rate (calculated)	≈ $0.21/hr
Filament	Spool weight / cost per gram (calculated)	985 g / ≈ $0.0152
Filament	AMS purge waste per color change	grams, to be estimated
Packing	Per box	3 min
Packing	Per item by class	Small 0.5, Standard 1, Multi-piece 1.5, Fragile 2 min
Packing	Pickup (no label) / counter sale	−1 min per box / 1 min per item
Ordering	Ordering cost per PO (S)	$15
Ordering	Service level (z)	95% (1.65)
Ordering	Periodic review interval	14 days
Holding	Carrying rates	See inventory section
Sales	Sales tax	Provo combined rate (verify before seeding); Utah orders only; shipping not taxed
Sales	Discounts	BYU 10%, business 15% on 25+ units, manual with reason
Sales	Returns	30 days; 10% restocking fee on changed-mind made-to-order
Quotes	Markup / minimum charge	2.5× / $10

Area	Setting	Default
Shipping	Rate table	Zones 1–8 by weight bracket (USPS Ground Advantage style)
Shipping	Transit days	2–5 by zone
Delivery	Runs to BYU / distance / mileage rate	Tue and Fri / 65 mi each way / $0.70 per mile
Capacity	Location capacities and warning levels	See inventory section
Products	Units per plate	1 (hints: Dragon 6, Spinner 4, Ornament 8)

Starting data and demand model

The system launches with 6 months of realistic history generated as real transactions, so every balance and report is derived, not typed.

Rubric minimum	Poulson Prints
25 items with barcode, cost, price, reorder point, lead time	47 items
5 parent items with BOMs	9
3 inventory locations	3 + QA Hold
6 suppliers with quoted lead time	7
10 customers	60 (11 with portal logins + 1 demo customer)
6 months of sales history	6 months back from launch date (about June–November 2026)

Demand model

$$\text{Daily demand} = \text{base} \times \text{trend} \times \text{season} \times \text{day of week} \times \text{events} \times \text{noise}$$

- **Base rates** per product (e.g. keychains about 6/day, dragons about 3/day, airplanes about 2/week).

- **Trend:** about 3% growth per month.
- **Season:** summer slow; late August–September back-to-school spike; November–December holiday ramp; ornament only in season.
- **Day of week:** online peaks Sunday/Monday evenings; campus counter weekdays only.
- **Events:** BYU home football games lift BYU Logo and blue items; Black Friday.
- **Color mix:** blue leads BYU items, white leads temples, black leads soldiers.
- **Channel:** about 80% online shipped, 20% campus counter or pickup.
- **Noise:** Poisson-style day-to-day variation, never flat or uniform random.
- The same model powers the **demand simulator** for live demo activity.

Customers — all fictional; emails use the reserved example.com domain (e.g. ethan.caldwell@example.com). 🔑 = portal login.

Type	Count	Buys	Pattern	Channel
BYU Student	18	Keychains, BYU Logo, spinners, dragons	Small, frequent; back-to-school and game-day spikes	Mostly campus pickup
Gift Buyer	16	Temples, magnet sets, airplanes	Larger orders; Nov–Dec ramp	Mostly shipped, many out of state
Hobbyist	12	Soldiers, airplanes, dragons, custom quotes	Steady, occasional big orders	Shipped
Local Family	9	Magnet sets, dragons, spinners, keychains	Summer and holiday bumps	Shipped and pickup
Business	5	Bulk keychains, spinners, dragons	Large, infrequent, Net 30	Shipped

#	Customer	Type	City	Portal
1	Ethan Caldwell	Student	Provo, UT	🔑
2	Madison Reeves	Student	Provo, UT	
3	Tyler Okafor	Student	Orem, UT	

#	Customer	Type	City	Portal
4	Hannah Lindqvist	Student	Provo, UT	🔑
5	Jacob Tuilagi	Student	Provo, UT	
6	Abigail Navarro	Student	Orem, UT	
7	Caleb Whitfield	Student	Provo, UT	
8	Emma Kawamoto	Student	Provo, UT	🔑
9	Samuel Brightwater	Student	Springville, UT	
10	Olivia Marchetti	Student	Provo, UT	
11	Noah Delacroix	Student	Orem, UT	
12	Grace Sorensen	Student	Provo, UT	🔑
13	Isaac Fonoti	Student	Provo, UT	
14	Lily Ashcroft	Student	Lehi, UT	
15	Daniel Park	Student	Provo, UT	
16	Chloe Vandermeer	Student	Provo, UT	
17	Mason Ellsworth	Student	Orem, UT	
18	Ava Thibodeaux	Student	Provo, UT	
19	Rebecca Holloway	Gift Buyer	Salt Lake City, UT	🔑
20	Mark Jensen	Gift Buyer	Sandy, UT	
21	Linda Castellanos	Gift Buyer	Mesa, AZ	🔑
22	Karen Whitaker	Gift Buyer	Boise, ID	
23	Steven Albright	Gift Buyer	Las Vegas, NV	
24	Rachel Nakamura	Gift Buyer	St. George, UT	
25	Paul Esposito	Gift Buyer	Sacramento, CA	
26	Julie Pemberton	Gift Buyer	Logan, UT	
27	Brian Oyelaran	Gift Buyer	Denver, CO	
28	Megan Halvorsen	Gift Buyer	Draper, UT	

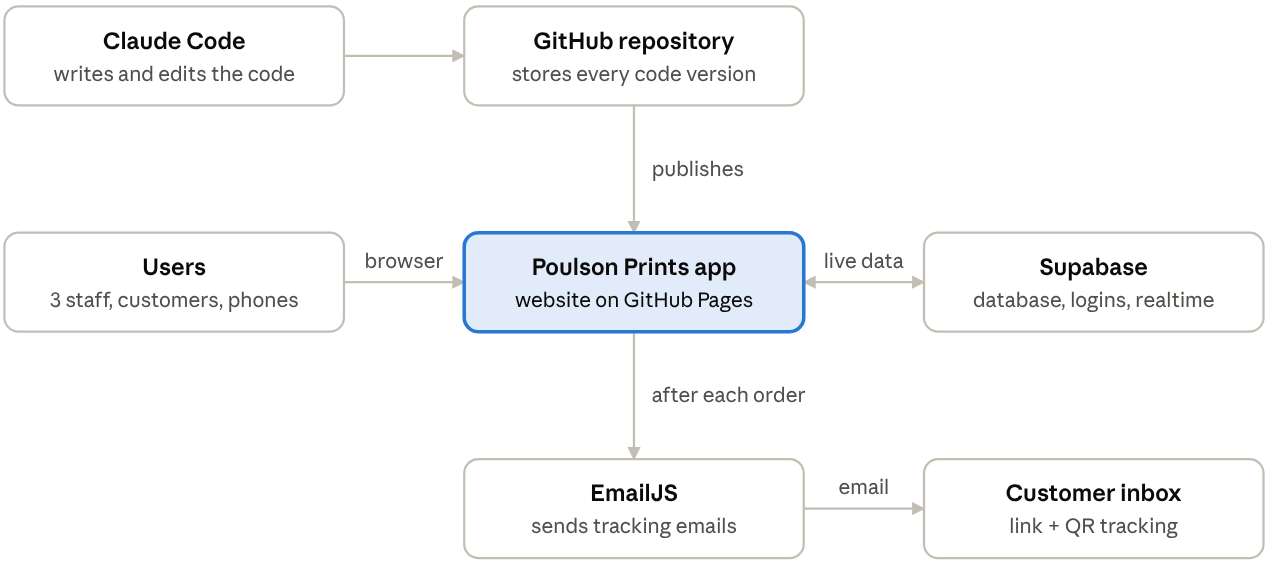
#	Customer	Type	City	Portal
29	Thomas Brennan	Gift Buyer	Idaho Falls, ID	
30	Susan Mortensen	Gift Buyer	Ogden, UT	
31	Gregory Lam	Gift Buyer	Seattle, WA	
32	Diane Fairbanks	Gift Buyer	Gilbert, AZ	
33	Kevin Andersen	Gift Buyer	Dallas, TX	
34	Natalie Crowther	Gift Buyer	Tooele, UT	
35	Marcus Bellweather	Hobbyist	West Jordan, UT	🔑
36	Derek Stroud	Hobbyist	Tooele, UT	
37	Aaron Velasquez	Hobbyist	Phoenix, AZ	
38	Brandon Kowalski	Hobbyist	Chicago, IL	
39	Jared Pettersen	Hobbyist	Grantsville, UT	
40	Victor Ramos	Hobbyist	Reno, NV	
41	Colton Hayes	Hobbyist	Stansbury Park, UT	
42	Wesley Chandler	Hobbyist	Portland, OR	
43	Trevor Lindgren	Hobbyist	American Fork, UT	
44	Dustin Moreau	Hobbyist	Austin, TX	
45	Kyle Fujimoto	Hobbyist	Layton, UT	
46	Russell Ngata	Hobbyist	Kearns, UT	
47	Jennifer Olsen	Local Family	Stansbury Park, UT	🔑
48	Amanda Christiansen	Local Family	Erda, UT	
49	Heather Larsen	Local Family	Tooele, UT	
50	Michelle Barlow	Local Family	Herriman, UT	

#	Customer	Type	City	Portal
51	Stephanie Gutierrez	Local Family	Spanish Fork, UT	
52	Laura McAllister	Local Family	Kaysville, UT	
53	Nicole Haddad	Local Family	Eagle Mountain, UT	
54	Crystal Pulsipher	Local Family	Cedar City, UT	
55	Angela Whitmore	Local Family	Saratoga Springs, UT	
56	Summit Ridge Gifts (Dana Moffat)	Business	Park City, UT	🔑
57	Lakeview Toy Chest (Ryan Castillo)	Business	Bountiful, UT	
58	Wasatch Tinkerers Guild (Jordan Ellis)	Business	Salt Lake City, UT	🔑
59	Canyon View Boutique (Priya Raman)	Business	Heber City, UT	
60	Provo Peak Party Supply (Ben Hartley)	Business	Provo, UT	🔑

Technical architecture

The stack follows Dr. Wright's recommended workflow (Claude Code → GitHub → GitHub Pages, data in Supabase), with all costs at $0.

The app runs on free services: code in GitHub, data in Supabase



system architecture · free services

Claude Code writes the app, GitHub stores it and publishes it as a website, and every user's browser reads and writes the same Supabase database in real time.

Piece	Choice	Why
Code writer	Claude Code	Writes all code; the team reviews and tests
Frontend	React + TypeScript (built with Vite)	Looks and behaves like commercial software
Database	Supabase (Postgres)	Shared data, logins, realtime updates, free tier
Hosting	GitHub Pages, built automatically by GitHub Actions on every push	Free, HTTPS (required for phone camera scanning)
Email	EmailJS + team Gmail	Free, sends to any address from a static site
Barcodes	Code 128 labels generated in the app; phone-camera scanning library	No extra hardware

- **All-or-nothing transactions:** sales, returns, receiving, production issue/complete and transfers run as database functions, so inventory and financials can never get out of sync.
- **Ledger table:** every transaction posts journal entries; the P&L, valuation and turnover are calculated from it.
- **Security:** the repository must be public for free GitHub Pages, so database access rules (row-level security) are required and the secret service-role key is never put in the code. The public "anon" key is safe to publish only with those rules on.
- **Documentation page:** the app lists Claude Code, Supabase, GitHub, GitHub Pages and EmailJS, as the rubric requires.

Setup guide for beginners

No coding experience is needed: you create the accounts and click a few settings, and Claude Code writes, connects and publishes the code. Plan about 1 hour for parts 1–3, done once by one teammate (the "builder") with the others watching.

Part 1 — Create the accounts (all free)

1. **Team Gmail.** Create one shared Gmail account for the project. Use it to sign up for everything below so no single teammate owns the system.
2. **GitHub** (stores the code and hosts the website). Go to github.com → Sign up with the team Gmail → choose the Free plan.
3. **Supabase** (the database and logins). Go to supabase.com → Start your project → Continue with GitHub. Signing in with GitHub links the two accounts.
    - Click New project. Name: poulson-prints. Region: a US West region. Plan: Free.
    - Click Generate a password for the database and save it in a password manager. Never paste it into code, chat or GitHub.
4. **EmailJS** (sends tracking emails). Go to emailjs.com → Sign up → Add New Service → Gmail → connect the team Gmail.

Part 2 — Install two tools on the builder's computer

1. **Git** (lets Claude Code send code to GitHub): download from git-scm.com, install with all default options.
2. **Node.js LTS** (runs the app on your computer for testing): download from nodejs.org, install with defaults.
3. Claude Code is already installed (this app). After installing, restart it.

Part 3 — Link everything (after you say "Begin coding", Claude Code walks you through each click)

1. **Create the repository.** On github.com click New repository. Name: poulson-prints-erp. Visibility: **Public** (free GitHub Pages requires it). Leave "Add a README" unchecked. Click Create, then copy the repository URL.
2. **Connect Claude Code to GitHub.** Paste the URL to Claude Code. It connects the project and pushes the code. The first push opens a browser window asking you to sign in to GitHub; approve it.
3. **Copy two Supabase values.** In Supabase open Project Settings → API. Copy the **Project URL** and the **anon / publishable key** (public). Do **not** copy the service_role / secret key.
4. **Store them in GitHub.** In the repository open Settings → Secrets and variables → Actions → New repository secret. Add the Supabase URL, the anon key and the three EmailJS values (Service ID, Template ID, Public Key) under the names Claude Code gives you.
5. **Turn on GitHub Pages.** Repository Settings → Pages → Source: **GitHub Actions**. Claude Code adds the file that builds and publishes the site on every push. Watch progress in the Actions tab; the site appears at https://(your-username).github.io/poulson-prints-erp/ within 1–2 minutes.
6. **Create the database.** Claude Code writes the database setup as SQL files. In Supabase open SQL Editor → New query, paste the file, click Run. Do the same for the starting-data file.
7. **Connect logins to the website.** In Supabase open Authentication → URL Configuration and set Site URL to your GitHub Pages address.
8. **Test.** Open the website, log in with a staff demo account and a customer demo account, and make one test sale.

Part 4 — Everyday workflow

1. Ask Claude Code for a change in plain English.
2. Claude Code runs the app on your computer so you can check it before publishing.
3. Claude Code saves a version (commit) and pushes to GitHub; the live site updates in 1–2 minutes.
4. If the change touched the database, re-run the updated SQL in the Supabase SQL Editor (Claude Code tells you when).
5. Only one teammate pushes at a time. Anyone else using Claude Code on their own computer pulls the latest version first.

Keep it safe and awake

- Never put the database password or the service_role key into code, chat or GitHub.
- Supabase pauses free projects after about a week with no activity. If the dashboard says paused, click Restore. Log in during the week before the demo.
- Keep all passwords and keys in one shared password manager.

Demo plan for 6 teammates

Each teammate owns two or three areas and backs up another, so whatever order Dr. Wright picks, someone who knows that screen deeply is ready. The demo is 10 minutes including questions; Dr. Wright chooses the scenario live.

Teammate	Owns (presents)	Backs up	Wow factor
Teammate 1 (name)	Point of Sale, Sales Orders, Customer portal	Returns	Live order tracking + email/QR
Teammate 2 (name)	Inventory Management, Barcode Support, capacity and holding costs	Production	Live spool tracking
Teammate 3 (name)	Returns and Exchanges, Financial Reporting	POS	Cost of poor quality on the P&L
Teammate 4 (name)	Production Orders, production report	Inventory	Print failure analytics + Report Failure
Teammate 5 (name)	MRP, Purchase Orders and Suppliers, EOQ/reorder points, scorecard	Financials	Capacity-capped EOQ
Teammate 6 (name)	Reporting, Multi-user logins, New Design wizard	MRP	Quote calculator + box selection

Every teammate should be able to explain one end-to-end story: a customer order flows to MRP and the printer queue, consumes filament and components, is packed and shipped, and posts revenue, COGS and fulfillment cost to the P&L.

Demo-day setup

- Two laptops side by side logged in as different staff users (proves multi-user), one logged in as the demo customer.

- A phone for camera scanning, plus printed barcode labels on real printed items and a spool.
- Reset demo data the night before; check Supabase is awake that morning.
- Requirements map page open in a tab to jump to any capability.
- Professional dress.

Build schedule (due Dec 2, 2026, 11:30pm)

1. Nov 11–17: accounts and setup, database, starting data, inventory, barcodes, POS, sales orders.
2. Nov 18–24: production, purchasing and suppliers, MRP, returns, financials, customer portal.
3. Nov 25–Dec 1: wow factors, polish, testing on several computers and phones, team practice runs (Thanksgiving week — plan around it).

Practice checklist

- [ ] Each teammate can demo their areas in under 90 seconds
- [ ] Each teammate can explain reorder point, EOQ and MRP netting in their own words
- [ ] Each teammate can change an input (e.g. lead time, ordering cost) and re-show the result
- [ ] Full run-through with someone else picking the order of capabilities
- [ ] Barcode labels printed and tested with the phone camera
- [ ] Tracking email tested to an outside address (and QR fallback)

Risks and open items

The biggest risk is a live-demo failure, which the rubric counts against the team unless the cause is external.

Risk	Mitigation
Supabase project paused on demo day	Log in during the final week; check that morning
Tracking email lands in spam or arrives late	QR code and link on the confirmation screen
Phone camera struggles to read a barcode	Good light, practiced scanning, typed-entry backup

Risk	Mitigation
Dr. Wright enters unexpected input (negative qty, oversell, over-receipt)	Validation messages, automatic backorders, over-receipt warning
Practice runs leave messy data	Reset demo data button
Teammates only know their own section	End-to-end story practice; everyone explains ROP, EOQ and MRP
Dates drift between build and demo	History generated relative to launch date; ornament in season in December

Open items

- [ ] Fill in teammate names in the demo plan
- [ ] Choose the team Gmail name
- [ ] Verify the Provo combined sales tax rate before seeding data
- [ ] Estimate AMS purge waste grams per color change
- [ ] Confirm the Hobby Lobby spool net weight on the box (985 g assumed)
- [ ] Team review of this spec, then say "Begin coding"